

B.Tech III Year II Semester (R19) Supplementary Examinations August 2023

DIGITAL COMPUTER PLATFORMS

(Electrical & Electronics Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Why Crystal is Preferred for a Clock Source? | 2M |
| (b) What is the program Counter? | 2M |
| (c) What is an instruction? | 2M |
| (d) What is Instruction cycle? | 2M |
| (e) Define micro controller. | 2M |
| (f) What is timer in microcontroller? | 2M |
| (g) List the scaling shifters available in C2xx DSP controller. | 2M |
| (h) Name the GPIO control registers of TMS320C2xx DSP. | 2M |
| (i) Define CPLD. | 2M |
| (j) List out the advantages of FPGA. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|---|----|
| 2 | (a) Briefly explain about the architecture of 8086 Microprocessor. | 5M |
| | (b) What are special function purpose registers in 8086 microprocessor. | 5M |
| OR | | |
| 3 | (a) Explain about the various flag registers in 8086 microprocessor. | 5M |
| | (b) Write about the various addressing modes of 8086 microprocessor. | 5M |
| 4 | (a) Explain the various modes of operation in 8255. | 5M |
| | (b) With a neat diagram explain the operation of 8259. | 5M |
| OR | | |
| 5 | (a) Explain the Need of DMA. | 5M |
| | (b) What are the various serial communication standards in 8086 microprocessor? | 5M |
| 6 | (a) Explain the TCON and TMOD registers of 8051 microcontroller. | 5M |
| | (b) Explain the various I/O instructions of 8051 microcontroller. | 5M |
| OR | | |
| 7 | (a) Describe the data storage scheme in a stack structure in 8051 microcontroller. | 5M |
| | (b) Explain the difference between forward jump and backward jump. | 5M |
| 8 | (a) Write TMS320LF2407 assembly language program to add two 16-bit data stored in the memory addresses 2000H and 2001H in the data memory and store the result in the data memory at addresses 3000H and 3001H. | 5M |
| | (b) Explain the indirect addressing mode used to access memory in TMS320LF2407. | 5M |
| OR | | |
| 9 | (a) Explain about the components of C2xx DSP core. | 5M |
| | (b) Discuss briefly about the physical memory available in TMS320LF2407 DSP. | 5M |
| 10 | (a) Explain the application of FPGA control in PWM controlled induction motor as a case study. | 5M |
| | (b) Explain the operations of FPGA based control implementation of DC-DC buck-boost converters. | 5M |
| OR | | |
| 11 | (a) Write about the programmable inter connect points. | 5M |
| | (b) Mention the feature of Virtex II pro FPGA boards. | 5M |
